Supplementary Material for: Sources of accuracy variation in deep crack-segmentation models: A controlled study of protocol, pretraining, budget, and priors

S1. Crack500 split units

Fig. S1 contrasts the two units of assignment used by the distributed Crack500 archive and by the parent-disjoint rebuild described in Section 2.2 of the main text.

S2. One TANGO propagation step

Fig. S3 expands one propagation step of the TANGO testbed (Section 2.4 of the main text): the tangent field assigns per-pixel softmax weights over four direction prototypes, the state passes through four line convolutions along those directions, and a learned gate admits the weighted sum into the state over three shared steps.

S2.1. Settings for reconstruction

This study's fourth contribution is a minimum reporting checklist, so the settings a reader would need to rebuild the testbed are given here rather than left to the released configuration. Orientation targets are computed from the ground-truth mask, not predicted: the mask is max-pooled to stride 4, smoothed with a Gaussian of $\sigma = 1.0$ over 5 taps, differentiated with Sobel operators, and the resulting structure tensor is smoothed again with $\sigma = 2.0$ over 9 taps; supervision is restricted to a 5-tap dilation band around the foreground, so background pixels carry no orientation loss. The propagation block uses 64 channels, four orientations, radius 5—eleven taps along each direction—and three shared steps, and the steering sharpness that multiplies the prototype responses before the softmax is a learned scalar initialized to 4.0. The main output is supervised by sigmoid cross-entropy with a foreground class weight of 5.0, a binary Dice term with smoothing 1.0, and a soft centerline-Dice term at five erosion iterations, weighted 1.0, 1.0 and 0.5; the orientation loss is weighted 0.5. One element of the training graph is absent from Fig. S2 and from the description above: a coarse evidence branch carries its own deep supervision, sigmoid cross-entropy at the same class weight and a Dice term, each weighted 0.4. The matrix row trains from random initialization. The two pretrained arms of Section 4.3 of the main text load, into the trunk and nothing else, the public SegNeXt MSCAN-S classification checkpoint `mscan_s_20230227-f33ccdf2.pth` for the ImageNet-1k arm, and the public SegNeXt segmentor trained on ADE20K, 44.16 mIoU, for the segmentation arm; the second matches the trunk on all 423 of its tensors by shape.

S3. Inference cost, training throughput and effective receptive fields

This section reports two ancillary measurements referred to in Sections 2.5, 4.1 and 4.5 of the main text.

S3.1. Inference latency and peak memory

The two measured-cost columns of Table 4 come from one sweep of all sixteen configurations in a single process on one idle NVIDIA RTX 4090 (WSL2, PyTorch 2.6.0+cu124, cuDNN autotuning disabled, TF32 left at the PyTorch defaults). Each model was built from its configuration alone, since weights do not change execution time, and its parameter count was recomputed and compared against Table 4 before timing, so a host holding a different architecture under the same name fails rather than contributing a number. The timed call is `encode_decode`, the entry point the GFLOPs column counts.

The protocol was chosen for reproducibility, not for resemblance to a deployment. At batch one—the unit the GFLOPs column counts—three back-to-back passes on the idle card disagreed by up to 46.1% for the lighter models and in both directions, and none of a longer warm-up, CUDA-event timing in place of a host clock, disabling cuDNN autotuning, or locking the graphics and memory clocks removed it; reporting the fastest iteration rather than the median did not either, as that statistic still moved by up to 53.7% across the same three passes. The card is time-shared with a Windows display stack, and the ratio of a model's median iteration to its fastest reaches 2.42 at batch one, which is consistent with a forward of many small kernels spending a variable share of its elapsed time waiting to be scheduled rather than computing. At batch eight that ratio falls to between 1.00 and 1.27, and the fastest of 80 timed iterations—taking contention to add time and never remove it—then reproduces: three independent passes agree to within 2.55%, and to better than 0.5% for fifteen of the sixteen models. The reported latency is that fastest iteration divided by eight; the reported memory is `max_memory_allocated` for the same forward, which excludes the CUDA context and the allocator reserve.

Two limits follow. The columns are one card and one software stack rather than a hardware-independent property, and they time an eager PyTorch forward rather than an export-and-optimize deployment path. They also depend on the fused kernels being present: SCSegamba's compatibility layer silently substitutes a pure-PyTorch selective scan when one `mamba_ssm`

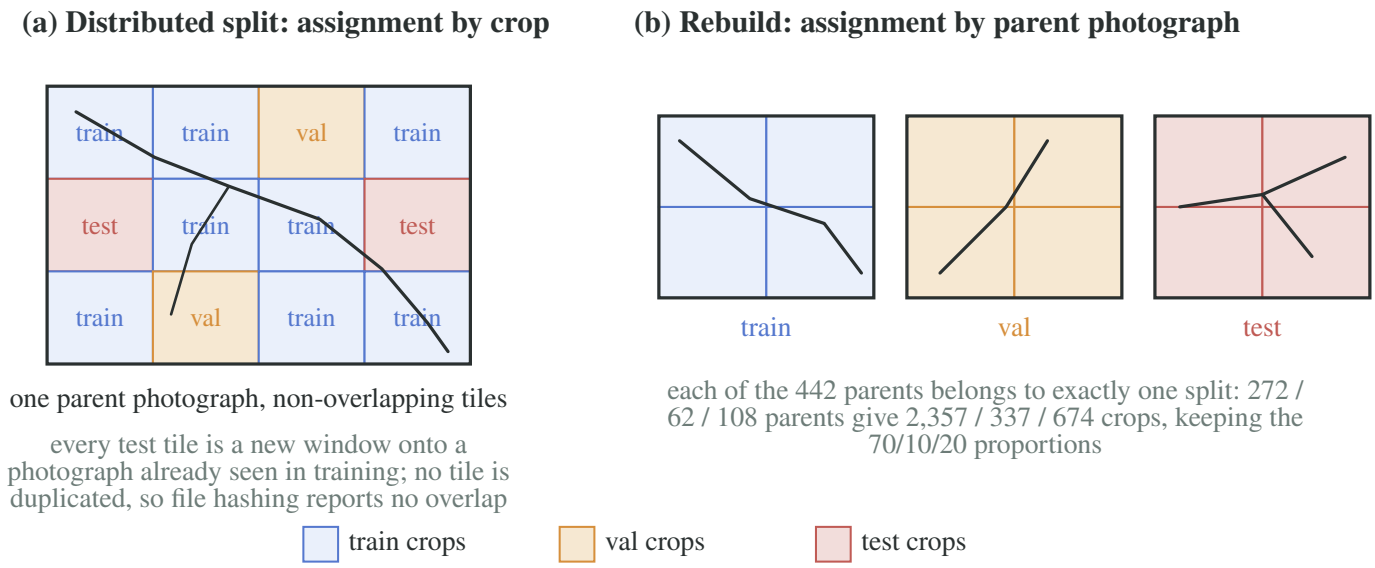


Fig. S1. Crack500 split units before and after the rebuild. (a) The distributed archive assigns non-overlapping tiles of one parent photograph to training, validation, and test, so every test tile is a new window onto a photograph already seen in training, while content hashing finds no duplicate tile. (b) The rebuild assigns each of the 442 parent photographs to exactly one split: 272/62/108 parents give 2,357/337/674 crops at the retained 70/10/20 proportions. Tile colors with in-tile labels mark split assignment; the crack polylines are schematic.

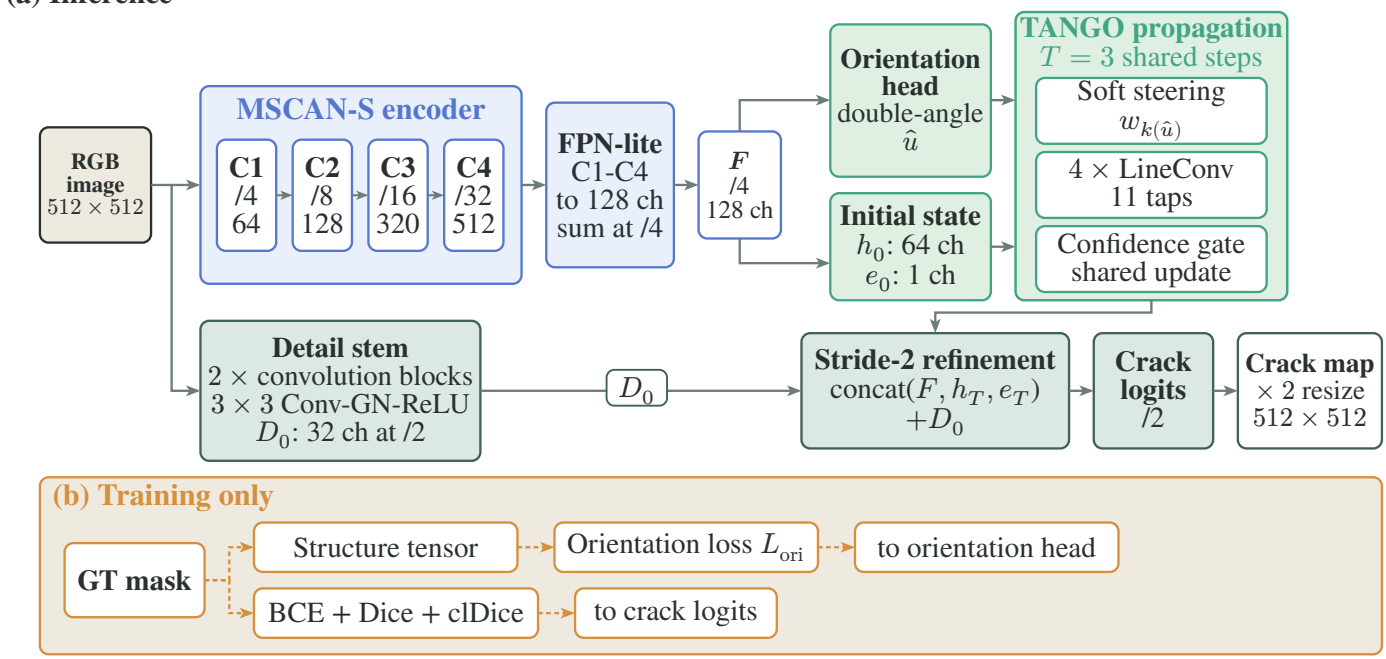


Fig. S2. TANGO testbed and independently switchable crack priors. Panel (a) shows the inference path with solid arrows; panel (b) shows training-only supervision with dashed amber arrows. h_T and e_T denote the propagation state and evidence channel after the $T = 3$ shared steps, $w_k(\hat{u})$ the per-pixel steering weights over the four line-convolution prototypes, and L_{ori} the orientation loss. In the layer labels, FPN is a feature pyramid network, BCE binary cross-entropy, GN group normalization, LineConv a line convolution, and ch a channel count; GT marks the ground-truth mask.

import fails, and on a host carrying a newer `mamba_ssm`, in which that module was renamed, the substitution ran about 4 s per single 512 by 512 forward against 56 ms with the real kernels, and no log reported the difference. The sweep aborts when it detects the substitution, and the reported run used the real kernels. This is the realized-value requirement of Table 7 applied to a cost measurement.

S3.2. Training throughput

Training throughput was audited only for TANGO and Crack-ShuttleNet—our local implementation based on the published description of ShuttleNet (Zhang et al., 2022)—on the parent-disjoint Crack500 split. The two runs matched hardware, software stack, input size, batch size, base optimizer, and 50-epoch exposure, and ran back to back on the same

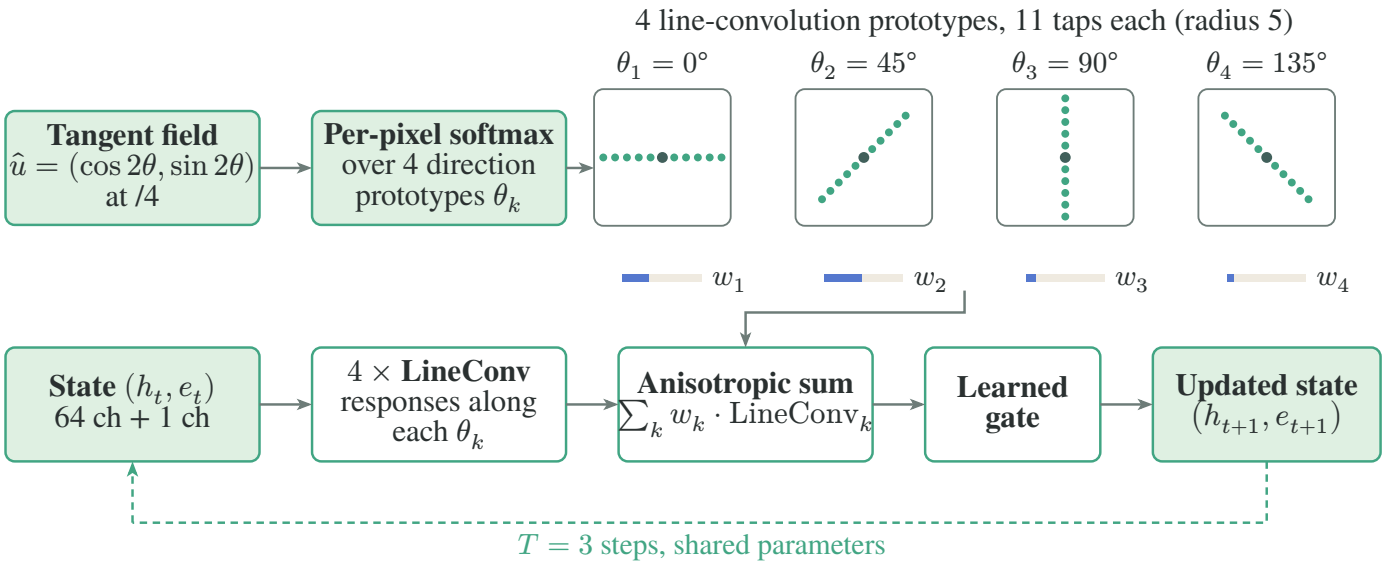


Fig. S3. One TANGO propagation step. The predicted tangent field sets per-pixel softmax weights w_k over four direction prototypes; the state passes through four 11-tap line convolutions (radius 5), whose responses are combined under those weights; a learned gate admits the combined update, and the three steps share one set of parameters. Bar heights over the kernel tiles illustrate the weights of one example pixel and are not measured values.

card with no other job on the machine; they did not match the optimizer wrapper, since TANGO trains under bfloat16 mixed precision with gradient clipping and Crack-ShuttleNet in single precision without (Table 2), two differences that move the per-iteration time in opposite directions. Mean seconds per iteration were calculated from 1,449 ten-iteration log windows each, after excluding the first warm-up window. Because only one matched run per model was available, these values are treated as implementation-level timing evidence rather than hardware-independent architectural properties.

Crack-ShuttleNet exceeded TANGO's four-dataset mean pixel-ODS by 1.17 points, 80.8 against 79.6, while requiring 49.5 instead of 23.9 billion counted operations for a 512 by 512 forward pass. TANGO's nominal operation count was 51.7% lower. The available matched timing run showed a different ordering: after warm-up, Crack-ShuttleNet averaged 0.264 seconds per iteration and TANGO 0.316 seconds on the same NVIDIA RTX 5090, making TANGO 19.6% slower in this implementation, computed from the unrounded per-iteration means. With one run per model, the timing is descriptive; Section S3.1 supplies the systematic version of the same point, measured for inference across all sixteen configurations.

S3.3. Effective receptive fields

Effective receptive fields were measured for four models on DeepCrack following Luo et al. (2016): the gradient of the crack logit at a probe pixel with respect to the input, with its magnitude averaged over the same 50 test images. Selective scan and attention make this gradient input-dependent, so the average over images is part of the measurement. Each field is summarized by r_{50} and r_{75}, the radii of the centered circles that contain 50 and 75 percent of the gradient mass.

Every field sits on a near-uniform floor. Normalization layers that aggregate over the spatial dimensions give each input pixel a small gradient irrespective of kernel or attention span, and the contribution of that floor to the accumulated mass grows with the square of the radius, so an uncorrected summary radius drifts outward with the quantile rather than with the reach of the model. The radii reported here are therefore measured after subtracting the floor, estimated for each field as the median of its radial profile between 220 and 250 pixels: outside every corrected r_{75}, and inside the largest centered circle that a 512 by 512 field samples completely. The correction is not cosmetic. Uncorrected, the four fields summarize as r_{75} = 41, 191, 151 and 165 pixels, and the last two of those measure the floor rather than the model: from 128 to 250 pixels the radial profile is flat for DTrC-Net and rises for MixerCSeg, while it falls by a factor of 8.4 for ConvNeXt-XLarge and 3.1 for TANGO. Quantiles above 75 percent are not reported, since the residual floor dominates them.

The probe sits at the geometric center of the image, which is a crack pixel in 5 of the 50 images; the median distance from the center to the nearest crack pixel is 34. What the measurement mostly describes is therefore the input dependence of a background prediction. Probing instead at the foreground pixel nearest the center, which 45 of the 50 images have within 128 pixels, and shifting each gradient map back to the center before averaging, gives r_{75} = 38, 143, 39 and 30 pixels for the four models, so the summary below is not an artifact of probing off the crack.

The average also runs over cracks at different orientations, so it describes extent and not orientation structure; it is evidence neither for nor against the anisotropy of any model. DTrC-Net processes a 256 by 256 resample of the input (Table 4), so its two radii describe that working domain rather than the 512 by 512 one.

The group labels in Table 4 of the main text record how each model builds global context, not how far the trained model

actually reaches. Fig. S4 shows the latter.

The measured extents do not follow the group labels. Three models drawn from three different groups land within two pixels of one another: ConvNeXt-XLarge at $r_{50} = 18$ and $r_{75} = 38$ pixels, the transformer DTrC-Net at 18 and 39, the Mamba model MixerCSeg faithful at 21 and 40. The one field that separates, $r_{50} = 87$ and $r_{75} = 140$, belongs to the remaining convolutional model, TANGO. The two models at opposite ends of the range are thus both convolutional.

Nor does that wide field come from the propagation TANGO is named for. Measured the same way on the arms of Table 6, removing the propagation gives $r_{75} = 140$ pixels and replacing anisotropic steering with uniform weights gives 143, against 141 for the reference arm trained with the same seed. Whatever sets the extent sits elsewhere in the network.

Extent also does not order the DeepCrack scores of these four models, measured on the seed-42 runs whose fields are shown, in pixel-ODS: a 3.7-fold difference in r_{75} accompanies a 2.0-point spread, and the narrowest and the widest field are the two highest-scoring of the four, 84.8 at 38 pixels and 84.9 at 140, the latter a single run above TANGO's three-seed mean of 84.5 (Table 5). Four models cannot establish a relationship between spatial extent and accuracy, and none is claimed here. The observation is narrower: the operator label does not predict the spatial support a trained model uses. Table 4 makes the same point for accuracy: without RINDNet the CNN group alone spans 77.9 to 81.6 mean held-out F1, a wider range than separates its best model from the best of either other group.

S4. Threshold rule: per-model detail, rank changes, and held-out operating points

The gain reported in Table 3 is non-negative by construction. Pooled foreground IoU and pooled F1 are related by $\text{IoU} = F_1/(2 - F_1)$, which is strictly increasing, so both are maximized at the same threshold and the value at t^* cannot fall below the value at 0.5. Across the 64 cells the two agree to within 0.0012 points. What that table reports is therefore the size of the loss, not its sign.

This section supplements Section 3.2 of the main text. Table S1 lists, for DeepCrack, the models that move two or more places when the threshold rule changes; Fig. S5 draws the fixed- and selected-threshold orderings model by model on DeepCrack and on distributed Crack500; and Table S2 reports the penalty of selecting the operating point on held-out data, together with the ranking by average precision.

On the five larger reported splits, held-out rank correlation is at least 0.997. Table S2 retains each split's penalties and both rank comparisons.

S4.1. Alternative shared thresholds

No single shared threshold preserves the model ordering. The sixteen DeepCrack curves admit no shared cut that leaves the ordering alone. The shared threshold that maximizes the mean F1 of the sixteen models is 0.235, and at that value RINDNet, whose own maximum lies at 0.697, falls to an F1 of 8.3, the value of an all-foreground prediction. The shared threshold that minimizes the worst loss among the fifteen models other than MixerCSeg faithful, whose scores never reach that region, is 0.383, and it still moves 12 of the 16 ranks and costs one model 1.46 points. The fixed cut at 0.5 is one member of this family, and for the typical model not its worst member. What the models do not share is a score scale, and no single threshold restores one.

S4.2. Score scales and training

Fixed-threshold losses occur on all four datasets (Table 3): the median foreground IoU gain does not exceed 1.43 points on any dataset, but a minority of models lose far more, most of all on DeepCrack. The size of the loss does not follow the distance between t^* and 0.5: the largest DeepCrack loss belongs to MixerCSeg faithful, whose t^* is 0.223, whereas DTrC-Net on the same dataset, whose t^* is 0.011 and so lies far further from the fixed cut, loses 1.36.

The extremes show what drives this: where each model places its scores (Supplementary Table S1; Fig. S6 traces three models' stored score maps at pixel level). MixerCSeg faithful, the published configuration, never places a score above 0.5 on two datasets it otherwise separates usefully: from the same predictions it measures 0.00 foreground IoU at 0.5 on DeepCrack and 70.83 at its selected threshold of 0.223, and on CrackMap 0.00 at 0.5 and 55.88 at its threshold of 0.419, so a fixed cut reports total failure in both cells. These are properties of where a model puts its scores, not of how well it ranks pixels, and of its training as much as its architecture: the released MixerCSeg weights, scored with this study's evaluator, select thresholds of 0.470, 0.506, 0.364 and 0.370 on the four datasets and lose 0.03, 0.00, 0.40 and 0.87 foreground-IoU points at the fixed cut (Section 2.3), whereas the same architecture retrained under the common recipe loses its entire foreground IoU on DeepCrack. The sensitivity is therefore a property of configurations trained under one shared recipe, of which the objective without foreground weighting is the most exposed (Table 2), not a constant of the architecture.

S4.3. Checkpoint selection at a fixed threshold

The same choice acts earlier, inside the training loop, where it selects which checkpoint becomes the result. The validation and test index files of the DeepCrack copy used here are identical (Section 2.2), so the best-checkpoint rule, foreground IoU at a threshold of 0.5, was applied to the test data; replacing the selected checkpoint with the prespecified final one measures what that selection contributed. Under the 50-epoch budget the two rules load different weights for all sixteen models, and selected-minus-final pixel-ODS has a median of +0.78 points and a maximum of +2.41. Most of that gain is not the leak: on the three datasets whose validation and test splits are disjoint, scored on the same partitions Table 4 reports, the same rule buys medians of +0.57, +0.18 and +0.39 points. The informative case runs the other way and is a threshold effect. MixerCSeg faithful selected epoch 4, which scores 47.96 foreground IoU at 0.5 against 0.00 for epoch 50; at each checkpoint's own operating

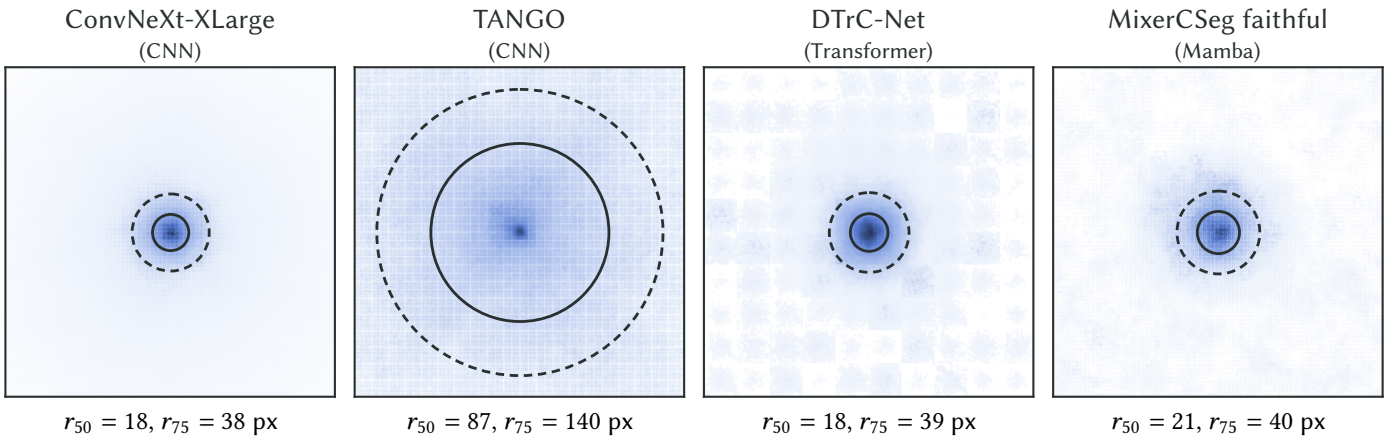


Fig. S4. Effective receptive fields of four models on DeepCrack. Each panel shows the magnitude of the gradient of the center-pixel crack logit with respect to the input, averaged over the same 50 test images, on a 322 by 322 center crop of the 512 by 512 input. Fields and radii are shown after subtracting the near-uniform floor described above. The solid and dashed circles mark r_{50} and r_{75}, the centered radii containing 50 and 75 percent of the remaining gradient mass; they are descriptive summaries, not the two-sigma Gaussian fit of the original method. Panels are normalized independently and shown under a fourth-root transform, so they compare spatial shape rather than gradient magnitude across models. The four models are the three case studies of Fig. S7 and the highest-scoring model of Table 4, covering the three operator groups; all use the prespecified final DeepCrack checkpoints.

Table S1. Representative models on DeepCrack: foreground IoU at a fixed and a selected threshold.

Model	t^*	**IoU at 0.5**	**IoU at t^***	**Rank 0.5**	**Rank t^***
TANGO (this study)	0.238	73.53	73.81	1	1
RINDNet (Pu et al., 2021)	0.697	71.50	71.71	2	7
Crack-ShuttleNet after Zhang et al. (2022)	0.127	70.84	71.56	6	9
CarNet (Li et al., 2024)	0.123	70.69	71.37	7	10
DTrC-Net after Xiang et al. (2023)	0.011	70.69	72.05	8	6
MixerCSeg modified after Zhao et al. (2026)	0.312	69.68	69.96	9	14
RestorMixer (Gu et al., 2025)	0.013	69.60	72.13	11	5
HRNet-W48 (Wang et al., 2021)	0.201	69.48	71.65	12	8
MixerCSeg faithful (Zhao et al., 2026)	0.223	0.00	70.83	16	12

Note: Ranks are over all 16 models of Table 4. Rows are the eight models that move two or more places on DeepCrack when the threshold rule changes, plus the highest-scoring model at 0.5; the lowest-scoring model at 0.5, MixerCSeg faithful, is already among the eight. Six further models change rank by one place (ConvNeXt-XLarge, SegFormer-B5, U-Net, DeepLabV3+ (R50), MambaVision, SCSegamba) and are not shown, so the fourteen changes counted in Table 3 cannot be read off this table. Cells are rounded to two decimals for display; gains quoted in the text are computed from the unrounded values. Each row carries its source under the model name; *after* marks a configuration derived locally from the cited architecture rather than the published model itself.

point the order reverses, 82.93 pixel-ODS against 70.10. Over fifty epochs the configuration became better at ranking pixels while moving its scores below the fixed cut, and the criterion read the improvement as a collapse: it chose the checkpoint that is 12.82 pixel-ODS points worse. Unlike the reporting effect above, this happens before any result is written down and cannot be corrected by rescoring, and it is the first of the two reasons for which every cell of Table 4 reports the prespecified

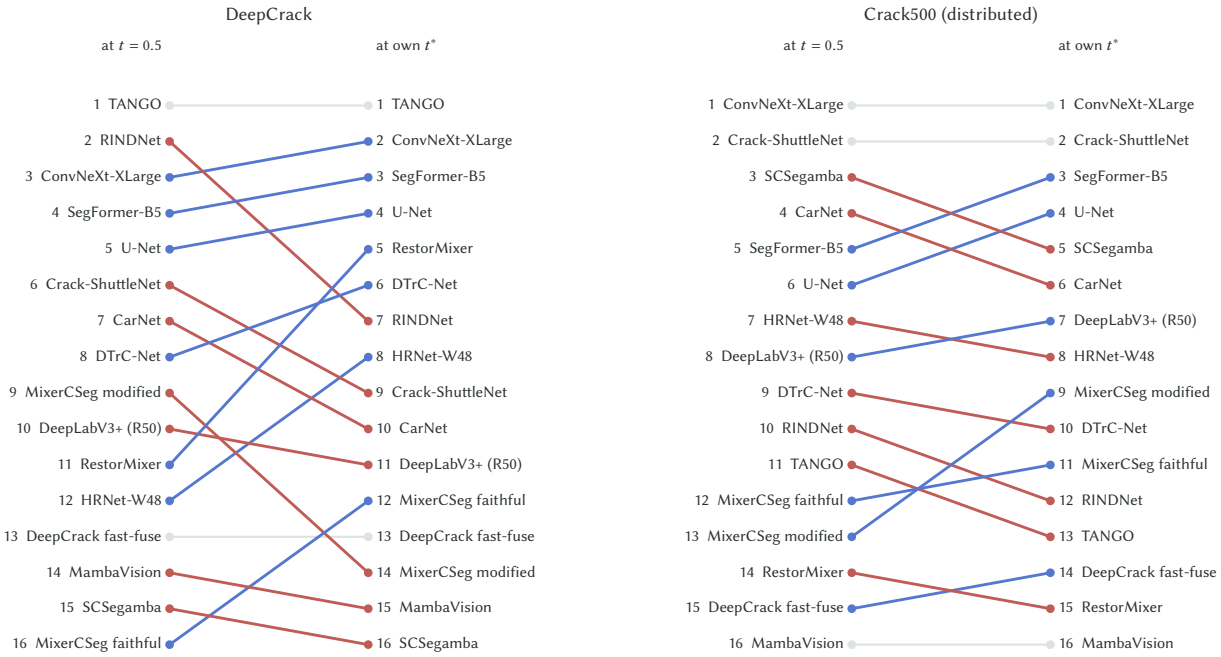


Fig. S5. Ranking at a fixed and a selected threshold, on DeepCrack and on distributed Crack500. Each line is one model, ordered by foreground IoU at $t = 0.5$ on the left of its panel and at that model's own dataset-level operating point t^* on the right; blue lines rise, red lines fall, and gray lines keep their position. Both orderings are computed from the same stored predictions, over all sixteen models of Table 4; no prediction differs between the two columns of either panel. CamCrack789 and CrackMap are summarized in Table 3 rather than drawn here.

Table S2. Selecting the operating point on held-out data, and ranking without one.

Reported split	**Images (t^* from)**	**Penalty median / worst**	**Ranks changed**	**Ranks changed by AP**	**ρ with pixel-ODS**
DeepCrack (half A)	119 (other half)	+0.02 / +0.14	0 of 16	14 of 16	0.59
DeepCrack (half B)	118 (other half)	+0.02 / +0.12	0 of 16	15 of 16	0.64
Crack500 (distributed)	675 (validation)	+0.03 / +0.07	0 of 16	14 of 16	0.67
Crack500 (p-disj.)	674 (validation)	+0.00 / +0.17	2 of 16	15 of 16	0.40
CamCrack789	157 (validation)	+0.02 / +0.25	2 of 16	16 of 16	0.56
CrackMap (s-disj.)	24 (validation)	+0.25 / +1.50	7 of 16	12 of 16	0.75

Note: Penalty is pixel-ODS at the test-selected t^* minus foreground F1 at the held-out t^*, in points; a positive value means the held-out choice scores lower. Ranks changed counts models whose position differs between the two orderings. AP is average precision over the same 1,000-bin threshold sweep, and ρ is the Spearman correlation between the AP ordering and the pixel-ODS ordering. The two DeepCrack rows are the two directions of the same split, made necessary by that dataset's duplicated validation list. CrackMap's selection split holds 12 images.

final checkpoint (Section 2.3).

S5. Qualitative case studies

Fig. S7 provides three model case studies, one sample from each of the four public datasets. The samples were prespecified from the input images and ground-truth masks before any prediction was viewed, and between them cover five conditions: alligator cracking on asphalt, a crack on concrete pavement, a thin linear crack on asphalt, and a crack whose contrast against its surroundings is the lowest in its dataset. Two are quantified on the inputs alone: the thin sample has the smallest mean crack width of the twenty-four CrackMap test images, 4.0 pixels, and the low-contrast sample has the smallest difference between the mean grey level inside the mask and in a 15-pixel ring around it, 0.056 of the intensity range, of the 237 DeepCrack test images. This morphology-first rule reduces selection based on favorable predictions.

TANGO represents the convolutional testbed, DTrC-Net a crack-specific Transformer comparator, and MixerCSeg faithful the published Mamba configuration, not the locally modified variant. These cases do not represent every member of their architecture group; all sixteen models remain in the quantitative comparison. What the panels show is read in Section 4.1 of the main text.

S5.1. Reading the main-text qualitative panel

Fig. S7 shows what that budget leaves. On the concrete crack of row S2-A the three models agree, recovering 90 to 96 percent of the marked pixels at their own operating points;

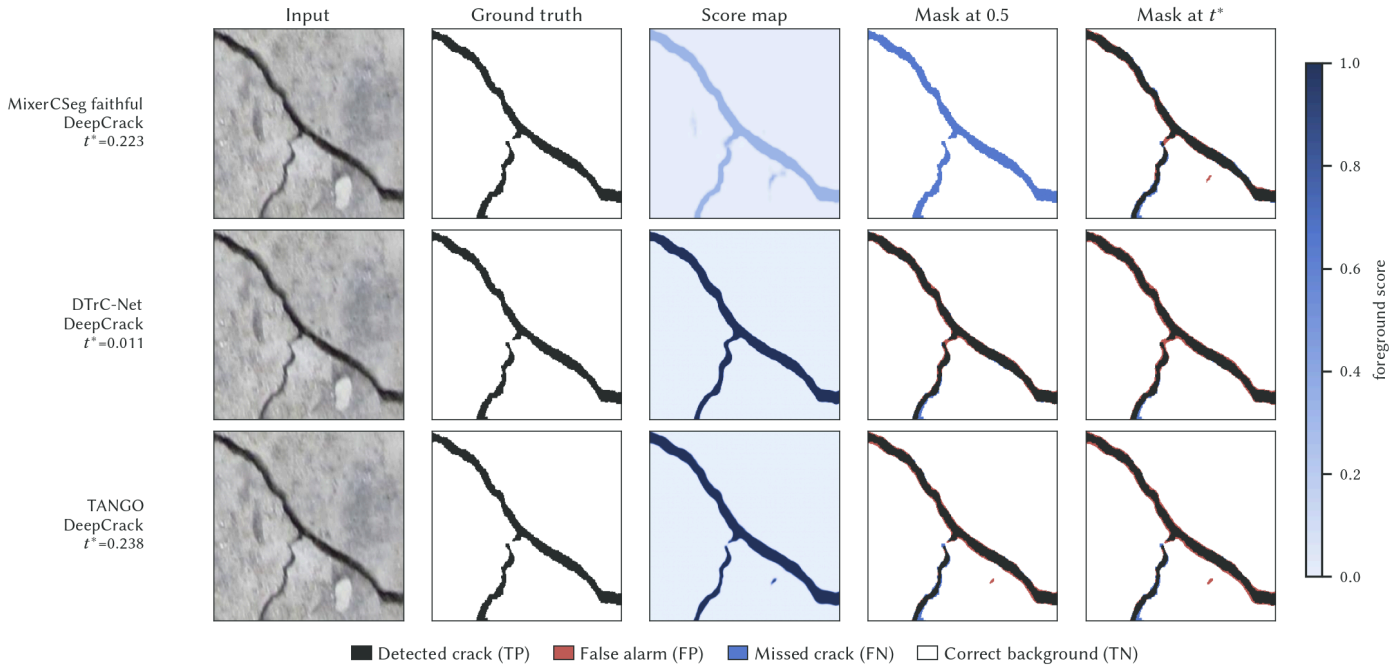


Fig. S6. One input, three models, two binarizations. Each row is one model on the same DeepCrack test image: the left panel thresholds at 0.5, the right at that model's dataset-level t^*, printed beneath it. Colors mark detected crack (TP), false alarm (FP), missed crack (FN) and correct background (TN), and the foreground IoU of each panel is printed on it. The three models are the case studies of Fig. S7. Section S4 gives the per-model sweep and the held-out operating points.

on the alligator pattern of row S1-A all three miss the finer branches, with recall between 50 and 66 percent. On the two harder rows they separate, and not in one direction: on the thin CrackMap crack of row S3-A DTrC-Net recovers 71 percent, TANGO 58 and MixerCSeg faithful 28, whereas on the low-contrast DeepCrack crack of row S4-A DTrC-Net recovers 7 percent, TANGO 44 and MixerCSeg faithful 47. A model that leads on one morphology trails on the next. Because architecture, morphology and dataset vary together across panels, the figure is an illustration; every attribution claim here rests on the tables.

S6. Training budget: seed spread and the whole-matrix arm

This section supplements Section 4.2 of the main text. Fig. S8 plots the eighteen runs of the three-seed arm (Table 5 of the main text), and Table S3 summarizes the whole-matrix arm.

S6.1. Initialization stratification

Averaged over the four datasets the pretrained six configurations gain 1.00 pixel-ODS points between the two budgets and the from-scratch ten gain 7.93, and 58 of the 60 pairings between the groups favour the from-scratch model. Removing the three collapsed cells, DeepCrack fast-fuse, MixerCSeg faithful and CarNet, all of them from-scratch, leaves 1.00 against 3.43. The ordering survives that removal—40 of the remaining 42 pairings still favour the from-scratch group—but the size does not, and the separation is uneven across datasets: every pairing favours the from-scratch group on CamCrack789, against 26 of 42 on DeepCrack, where the difference after those removals is 1.16 points.

The pretrained group leads the from-scratch group by 8.54 points at 5 epochs and by 1.61 at 50, a gap five times larger at the short budget. This is a descriptive comparison of the same configurations at two budgets, not an independent matched experiment.

S6.2. Limits of the initialization stratification

Two limits apply to the initialization stratification reported in Section 4.2 of the main text. Each cell is one run, so the group difference carries seed variation and is not a matched contrast; the repeated-run evidence remains the three-architecture arm. The groups are also not randomly assigned. The pretrained group is the heavier one, at a median of 135.1 against 36.7 GFLOPs and 62.6 against 9.6 million parameters, and a median split of the sixteen configurations by parameter count separates the gains almost as cleanly as the initialization label does: 51 of the 64 cross-half pairings favour the lighter half, against 58 of 60 for the initialization split, and 40 of 64 when the split is by counted operations. U-Net, whose 6.29-point gain and rise from eleventh place to third ties CarNet for the largest move in the matrix in either direction, is heavier than four of the six pretrained configurations by counted operations but lighter than all six by parameter count, so it does not decide between the two readings. The stratification therefore shows that the short budget records initialization, capacity or both; it cannot apportion the gain between them.

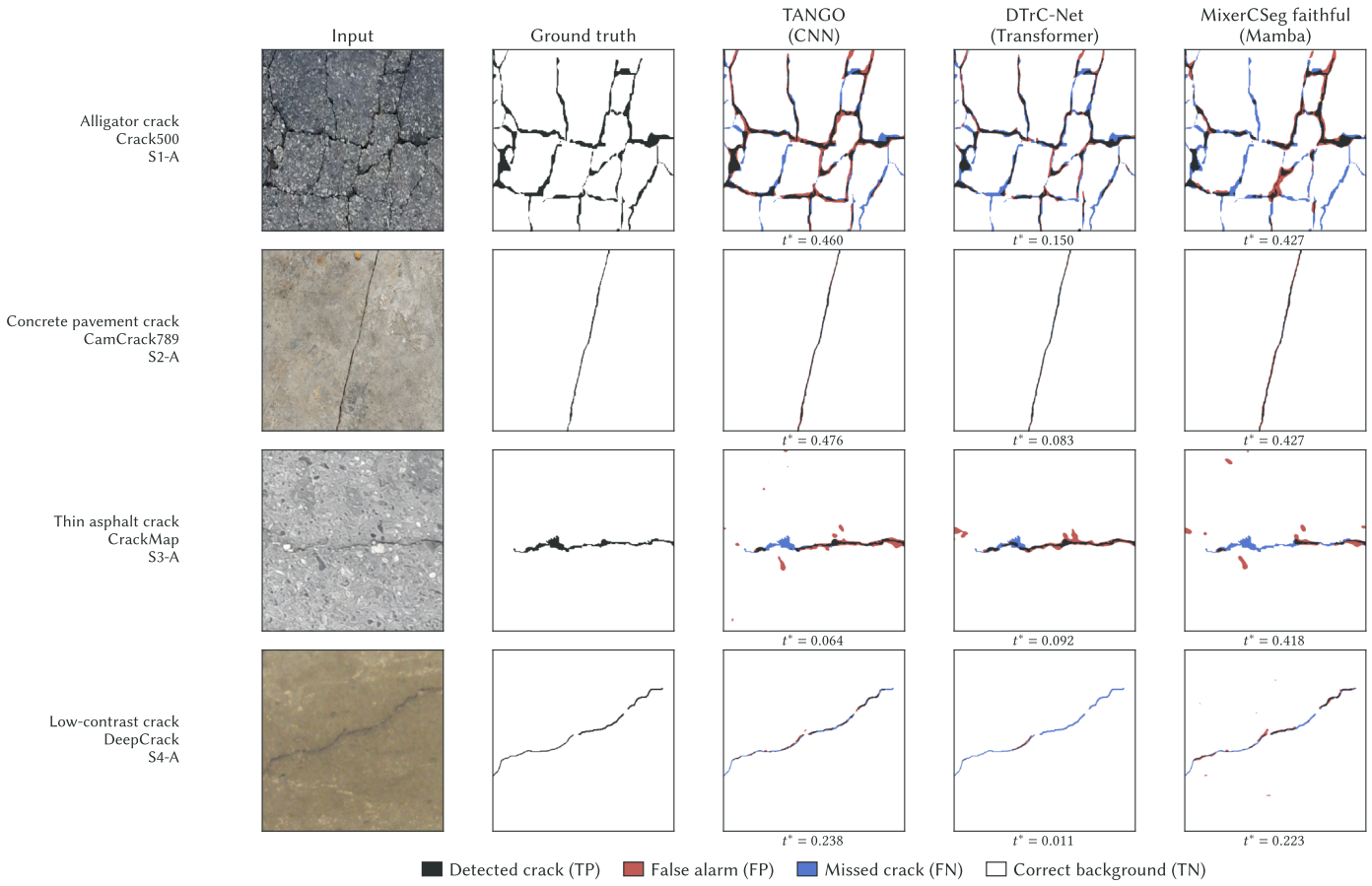


Fig. S7. Qualitative comparison, one sample from each of the four public datasets: an asphalt alligator crack, a concrete pavement crack, a thin asphalt crack, and the lowest-contrast crack in its dataset (rows S1-A to S4-A; row S1-A uses the distributed Crack500 partition). TANGO, DTrC-Net and MixerCSeg faithful stand in for the convolutional, Transformer and Mamba groups. Predictions use each model's dataset-level pixel-ODS threshold, printed as t^* beneath the panel it applies to; the legend distinguishes detected crack (TP), false alarm (FP), missed crack (FN) and correct background (TN). Samples were fixed from inputs and annotations before any prediction was viewed, by the morphology-first rule of Section S5.

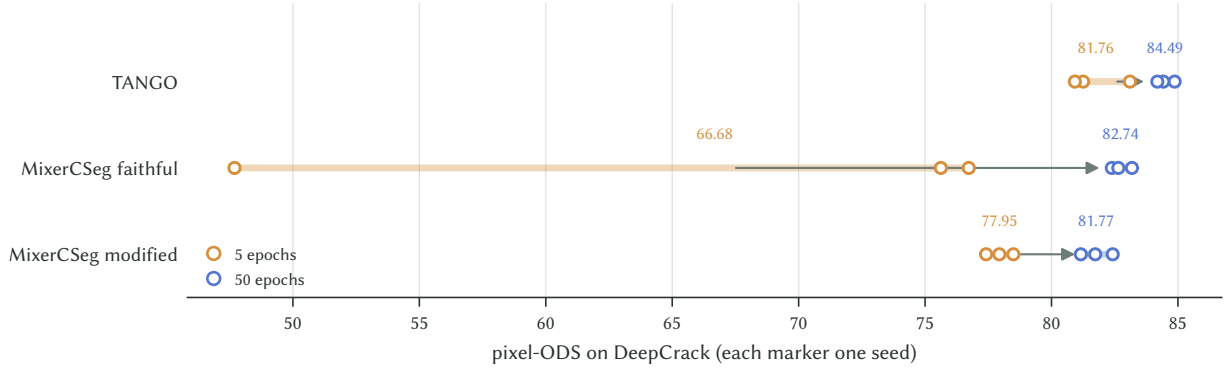


Fig. S8. Training budget and seed spread at fixed architecture. Each marker is one run's pixel-ODS on DeepCrack; amber marks 5 epochs, blue 50 epochs, and each arrow connects the two budget means of one configuration. The values are those of Table 5: every mean rises under the tenfold budget, and the across-seed range narrows for two of the three configurations.

S6.3. The matched pretraining contrast

The stratification above cannot separate initialization from capacity, because its two groups differ in model family and scale as well as in starting weights. The contrast reported here can. It holds the architecture, the data and the 50-epoch protocol fixed and changes only the weights loaded into the MSCAN-S trunk: random, the trunk of the public SegNeXt segmentor trained on ADE20K, or the ImageNet-1k classification checkpoint of the same trunk. All three arms were run on four datasets with seeds 42 to 44 in one queue on one machine, thirty-six runs, so both priors are measured against the same from-scratch runs (Table S4). An earlier version of this study

Table S3. Whole-matrix budget contrast: the same sixteen configurations at 5 and 50 epochs, one seed per cell (pixel-ODS).

Dataset	Updates at 50 ep	Spread 5 ep	Spread 50 ep	Mean change	Largest change	Ranks changed
CrackMap	550	56.96	17.99	+10.73	+55.01	14 of 16
DeepCrack	1,900	20.42	5.21	+5.16	+20.45	16 of 16
CamCrack789	3,450	24.60	5.22	+5.28	+19.69	13 of 16
Crack500 (dist.)	14,700	6.42	10.16	+0.35	+3.08	15 of 16

Note: Rows are ordered by the number of optimizer updates the 50-epoch budget supplies, which is what an epoch count converts into and which differs 27-fold across these datasets. Spread is the highest minus the lowest of the sixteen pixel-ODS values on that dataset. Mean and largest change are over the sixteen per-model differences, 50 epochs minus 5. Ranks changed counts the models whose position among the sixteen differs between the two budgets; the Spearman rank correlation between the two orderings is 0.33, 0.61, 0.83 and 0.51 in row order. Every model improves on DeepCrack, fifteen of sixteen on CamCrack789, thirteen of sixteen on CrackMap, and eight of sixteen on Crack500. Each cell is a single run, so the per-model differences carry seed variation; Table 5 supplies the repeated-run estimate of that variation. RINDNet additionally changed micro-batch between the two arms (Section 2.3).

compared each prior with its own separately trained reference, which made the two effects impossible to rank; the shared reference removes that problem.

The two priors are indistinguishable on every dataset, and both separate from the reference on Crack500 and CamCrack789. CrackMap's gains are the largest but its runs vary most, and DeepCrack's do not separate. Validation foreground IoU, recorded every five epochs at the fixed threshold of 0.5, shows how the gains arise. On DeepCrack the pretrained arms lead the reference by 4.94 (ImageNet-1k) and 6.37 points (ADE20K) after five epochs and by 0.27 and 0.80 after fifty, a convergence advantage that the longer budget erases. On the other three datasets the lead after fifty epochs is about as large as after five, 1.10 to 8.44 points against 0.41 to 6.82, so pretraining there yields a better endpoint, not only an earlier one.

Two of these datasets were rescored after the split repair of Section S1, and the repair changed the contrast itself, not only its level. On Crack500 the difference moved from +1.45 to +1.37 pixel-ODS. On CrackMap it moved from +6.10 to +3.22, and almost all of that came from one arm: removing the parent images of the test set from training cost the pretrained arm 2.44 points, while the from-scratch arm gained 0.45. Leakage there did not lift the two arms alike, it rewarded the arm that arrived with a prior. An initialization study run on the released CrackMap split would therefore have reported an effect nearly twice the size of the one measured here, and a leaked split cannot be treated as merely optimistic about the level while still trustworthy about the comparison.

S6.4. Budget and the fixed-threshold penalty

For TANGO and MixerCSeg modified the penalty imposed by the fixed threshold shrinks as the budget grows (Section 4.2 of the main text reports MixerCSeg faithful). The gap between mean foreground IoU at the selected operating point and at 0.5 falls from 2.98 points at 5 epochs to 0.39 at 50 for MixerCSeg modified, and from 0.75 to 0.39 for TANGO. The whole-matrix arm agrees for the typical model: the median gain of Table 3 is 1.43, 0.29, 0.47 and 0.92 points at 50 epochs against 2.45, 0.79, 1.04 and 3.30 at 5. It disagrees for the extreme. MixerCSeg faithful's DeepCrack gain grows from 6.59 to 70.83 points between the two budgets, because over the longer run it moved its scores entirely below the fixed cut while becoming better at ranking pixels. A longer budget makes a fixed threshold safer for the median model and can make it catastrophic for an individual one, so neither budget makes the operating point safe to leave unstated.

The from-scratch TANGO reference of Table 6 scores 84.51 mean held-out F1 over its five seeds on DeepCrack, with an across-seed range of 0.69, and 77.16 on parent-disjoint Crack500, with a range of 0.18; its centerline Dice is 88.27 (range 0.49) and 80.32 (range 0.44). Means and ranges are computed from the unrounded results. Table S9 lists every run behind that table and the two p values of each contrast. The submitted version marked a contrast whenever the mean of its three paired changes exceeded their range; that rule is not a test at a stated level, since with a true change of zero it would still mark a contrast about 8% of the time at three seeds, and it is replaced here by paired t intervals with Holm correction.

S6.5. Repeated runs and the optimizer wrapper

Every architecture improves, but the increases differ by a factor of six, from 2.73 points for TANGO to 16.06 for MixerCSeg faithful (Table 5). The two budgets differ in more than the number of updates: because the schedule is defined over the epoch budget (Section 2.3), its floor occupies one fifth of the short run and one fiftieth of the long one, so part of each increase comes from a larger share of the budget spent at a useful learning rate. The contrast is between two complete recipes at two budgets, which is how an epoch count is chosen in practice. The budget also changes which MixerCSeg configuration leads: the modified variant is ahead by 11.27 points at 5 epochs, whereas at 50 epochs the two are within one point with overlapping seed ranges. A shared epoch budget can place architectures at different stages of optimization.

Across three seeds, the tenfold budget narrows the observed range for TANGO and MixerCSeg faithful, by factors of 3.2 and 36.4, and widens it slightly for MixerCSeg modified (Table 5). At the short budget, MixerCSeg faithful spans 29.04 points across seeds, one seed reaching 47.69 while the other two land near 76, far larger than any between-model difference in the 50-epoch matrix; a single run of that configuration reports the seed as much as the model. The corresponding cell

Table S4. Pretraining source against one shared from-scratch reference: the same TANGO trunk, four datasets, seeds 42, 43, 44, held-out F1. The change is the pretrained arm minus the reference, or ImageNet-1k minus ADE20K, paired by seed, with its 95% paired t interval; bold marks a Holm-adjusted p below 0.05 within the twelve contrasts.

Dataset	Initialization	Per-seed F1	Mean	Change [95% CI]	Holm p
CrackMap (p-disj.)	from scratch	72.12 / 71.34 / 71.97	71.81		
	ADE20K	74.50 / 74.59 / 75.02	74.70	+2.89 [+1.76, +4.02]	0.065
	ImageNet-1k	74.33 / 73.98 / 75.84	74.72	+2.91 [+0.77, +5.04]	0.196
	IN-1k minus ADE20K			+0.02 [−1.80, +1.84]	1.000
CamCrack789	from scratch	84.57 / 84.59 / 84.59	84.58		
	ADE20K	86.28 / 86.37 / 86.67	86.44	**+1.85** [+1.37, +2.33]	0.033
	ImageNet-1k	86.45 / 86.25 / 86.48	86.39	**+1.81** [+1.49, +2.13]	0.018
	IN-1k minus ADE20K			−0.04 [−0.53, +0.44]	1.000
Crack500 (p-disj.)	from scratch	77.16 / 77.18 / 77.20	77.18		
	ADE20K	78.56 / 78.51 / 78.55	78.54	**+1.36** [+1.27, +1.45]	0.003
	ImageNet-1k	78.39 / 78.32 / 78.53	78.41	**+1.23** [+1.01, +1.45]	0.018
	IN-1k minus ADE20K			−0.13 [−0.34, +0.09]	0.756
DeepCrack	from scratch	84.98 / 84.12 / 84.41	84.50		
	ADE20K	85.90 / 84.92 / 84.49	85.11	+0.60 [−0.52, +1.73]	0.756
	ImageNet-1k	84.96 / 84.63 / 84.47	84.69	+0.18 [−0.53, +0.90]	1.000
	IN-1k minus ADE20K			−0.42 [−1.60, +0.76]	1.000

of Table 4 was trained under the matrix recipe, without the gradient clipping and two-epoch warm-up of the three-seed arm (Table 2). Repeated at three seeds under that recipe, MixerCSeg faithful averages 71.68 pixel-ODS at 5 epochs over a range of 14.22 and 81.95 over 1.17 at 50: the short-budget collapse is milder and less seed-dependent without clipping, but its range still exceeds any between-model difference in the 50-epoch matrix, and the matrix cells sit at the edges of these ranges rather than inside them.

S6.6. TANGO without clipping

Gradient clipping is present in exactly the two configurations this study wrote (Table 2) and is the standard remedy for the collapse just described, so the stability reported for TANGO could be a property of its wrapper rather than of the architecture. Retraining TANGO at three seeds under the twelve baselines' wrapper—single precision, no clipping, no weight-decay grouping—separates the two. It averages 84.51 pixel-ODS at 50 epochs over a range of 1.08, against 84.49 over 0.68 with the original wrapper, a paired difference of +0.02 points that changes sign across seeds; at 5 epochs the figures are 81.95 over 1.27 against 81.76 over 2.18, a paired difference of +0.19, again sign-changing. The budget effect is +2.57 points under the baselines' wrapper and +2.73 under the original one. TANGO trains as stably without clipping as with it on this dataset, so the seed ranges reported for it are not an artifact of the wrapper; what the wrapper does explain is part of the short-budget behaviour of MixerCSeg faithful, which is the configuration those ranges are contrasted against.

These runs also cross-check the threshold finding of Section 3.2 on independent checkpoints: MixerCSeg faithful placed no score above 0.5 in five of its six runs, so the fixed threshold is uninformative for it at both budgets while its operating-point scores are not.

S6.7. Whole-matrix behaviour and short-budget collapse

The whole-matrix arm shows what that variation does to a benchmark table. Cutting the budget tenfold widens the between-model spread on three of the four datasets: from 5.21 to 20.42 pixel-ODS points on DeepCrack, from 5.22 to 24.60 on CamCrack789, and from 17.99 to 56.96 on CrackMap. On Crack500 it does the opposite, narrowing the spread from 10.16 to 6.42.

The four datasets order by the number of optimizer updates their training sets supply, not by any property of their imagery, and the budget effect follows that order. CrackMap, whose 84 training images yield about 550 updates at 50 epochs and 55 at 5, gains 10.73 points on average and 55.01 in its largest cell. Crack500, whose 2,355 crops already supply 1,470 updates at the short budget, gains 0.35 points on average, and eight of its sixteen cells score lower at the longer budget. On the three datasets that gain, the gain is almost exactly the distance from a common 50-epoch value: the between-model standard deviation falls from 5.68 to 1.39 pixel-ODS points on DeepCrack, 6.44 to 1.29 on CamCrack789 and 18.33 to 3.99 on CrackMap, while on Crack500 it does not fall at all, 1.93 to 2.37. Correlating a configuration's short-budget score against its own change adds nothing, because for two independently varying arms that correlation is fixed by the two spreads alone; what the two arms' correlation does report is whether a configuration keeps its standing: 0.92 between the budgets on CamCrack789 but 0.44 on DeepCrack and 0.10 on CrackMap, so where the collapse is largest a 50-epoch score is mostly not predicted by a 5-epoch one.

The three cells that gain most are three configurations on the smallest dataset: DeepCrack fast-fuse from 17.27 to 72.27, MixerCSeg faithful from 21.59 to 71.69, and CarNet from 42.88 to 74.65. At the short budget these are also the three lowest cells of the whole matrix, and rescoring cannot recover them, since they are already measured at their maximizing thresholds. Repeating the DeepCrack fast-fuse cell at three

seeds gives 15.3, 8.0 and 8.0 pixel-ODS, and its per-epoch validation foreground IoU at 0.5 never exceeds 4 while the training loss falls smoothly: the signature of a model held at the all-background solution, not of unstable optimization (its published recipe trains for 500 epochs). All three train without gradient clipping (Table 2), and MixerCSeg faithful, whose objective carries no foreground weighting, is also the configuration behind the largest threshold loss and the selection loss of Section 3.2; what the objective and the wrapper contribute is not separated here, though the control arm above separates the wrapper for TANGO.

S7. Capacity: the fuse-order pair

One comparison in this matrix comes close to isolating computation. The two DeepCrack side-fusion orders described with Table 4 have identical parameter counts, tensor for tensor, and differ in architecture only in where the fusion convolution sits relative to the upsample. At the scale-8 and scale-16 sides, convolving after the upsample rather than before it puts 64 and 256 times as many pixels through that convolution, and the published order accordingly counts 549.8 GFLOPs against 202.4, a factor of 2.72 at the same 30.9 million parameters. Retrained under the identical protocol and rescored with the same evaluator, the published order reaches 82.2, 75.0, 82.9 and 71.9 pixel-ODS against 82.5, 75.1, 83.4 and 72.3 for the row of Table 4, taking Crack500 from the parent-disjoint column throughout: a four-dataset mean of 78.0 against 78.3, the Mean cell of that row. No per-dataset difference exceeds 0.49 points, and the 0.29-point mean difference is smaller than every across-seed range in Table 5, which span 0.68 to 1.26 points at this budget. With one seed per cell the two orders are therefore not distinguishable in accuracy here; what is not in doubt is that one of them costs 2.72 times the computation.

The published order does not fit the common protocol, and that is what stops the pair from being a single-variable contrast. At 512 by 512 and an effective batch of 8 it fails with a CUDA error, so it ran at micro-batches of 2 with four-step accumulation, taking 278 minutes against 61 for the same 50 epochs on the same data and card. This implementation divides each side loss by the *physical* batch, 8 against 2, while the optimizer wrapper rescales an accumulated step only by the accumulation count, 4. The published order therefore trains at four times the effective loss scale of the row in Table 4—an optimization difference, not an architectural one, and exactly the kind of unstandardized wrapper setting Section 4.2 reports for two other configurations (Table 2). The accuracy comparison accordingly does not isolate fusion order and is not offered as a within-configuration result. What the pair does establish is the cost: at identical parameter counts one order puts 2.72 times as many operations through the same convolutions and takes 4.6 times as long to train, and the accuracy it buys is not resolved here in either direction.

S8. Protocol, evaluator, and matrix details

This section holds implementation-level detail referred to in Sections 2.3, 2.5 and 4.1 of the main text.

S8.1. Implementation defects

The realized learning rates of the common schedule fall from 0.000500 in the first epoch to 0.000011 in the forty-ninth and 0.000001 in the fiftieth, which runs 500 times below the initial rate. The three configurations trained at micro-batches of 2 also used two dataloader workers instead of four.

The declared initial learning rate of MixerCSeg modified was 5×10^{-4}, but its logged values stayed between about 5×10^{-7} and 1×10^{-6}, roughly 500 to 1,000 times smaller, and the run completed without a warning. The cause was a warmup window of one epoch: under epoch-based scheduling one epoch supplies a single interpolation step, so the warmup never reached its end factor and left the rate at the product of the start factor and the base rate, which the following polynomial decay then inherited in place of the declared value. No other configuration used a warmup shorter than its decay schedule.

In the group-normalization defect, the eight-group implementation and the published one-group-per-eight-channels implementation agree only where a layer has 64 channels. Module structure and parameter names were unaffected, so the released weights loaded into the 180-bin evaluation configuration with no missing or mismatched tensor, and the defect became visible only when both implementations were run on those weights.

The scheduler was corrected by widening the warm-up window to two epochs. Rerunning the affected configuration changed pixel-ODS by 9.96 to 69.99 points across datasets. The normalization correction raised the correlation between the two implementations' logits from 0.40 to 0.996. Both MixerCSeg rows were retrained after that correction; all reported values use the corrected runs.

S8.2. Released MixerCSeg weights

Loaded into the corrected implementation, the released weights reach 87.43, 77.56, 82.06 and 78.30 pixel-ODS on DeepCrack, Crack500, CamCrack789 and CrackMap. These are not the values published with them, which were measured on the authors' own random re-partition of each dataset with an image-averaged variant of ODS; under the protocol used here the published model scores above its published value on three of the four datasets and 3.5 points below on DeepCrack. The protocol therefore does not by itself account for the distance between the released weights and the retrained MixerCSeg faithful row of Table 4, which is 4.50, 0.55, 0.83 and 7.61 points on the four datasets. Training budget accounts for part of it: the released checkpoints are stored at epochs 15 to 28 of a 50-epoch run at batch size one, so they carry 2.4 to 4.5 times as many optimizer updates as the 50-epoch, batch-8 protocol used here, by the same factor on every dataset. A constant shortfall in updates costs most where the absolute count is smallest, which is what the four gaps show: 0.55 and 0.83 points on the two larger datasets against 4.50 and 7.61 on the two smaller ones. The two also differ in one architectural setting, which the

comparison cannot separate from budget: the released weights use 180 histogram-of-gradients bins, whereas the released code, and therefore the retrained row, uses 36, a difference of 34,560 parameters and the only structural one.

S8.3. Evaluator fields, determinism, and operation counts

The evaluator stores an auxiliary field named pixel-OIS, obtained by selecting a threshold per image and then micro-aggregating the resulting counts. That implementation can produce a value below pixel-ODS, so it is not interpreted as a formal upper bound and is not reported. Mean IoU is likewise stored but not used. The artifact holds nine metric values per model in total: pixel-ODS together with the precision and recall from which it is formed, foreground IoU and mean IoU at both t^* and $t = 0.5$, pixel-OIS, and centerline Dice. Two evaluation passes over the same sixteen DeepCrack checkpoints, run as separate processes, returned identical values for all nine stored metrics, so the evaluator itself is deterministic.

Operators the analyzer cannot count, including fused attention, selective scan, and some convolution variants, are marked as lower bounds, not treated as complete operation counts; the released artifact lists the uncounted operators for each such row. The analyzer excludes elementwise, normalization, activation and pooling operators. Two models were each measured on two different machines; the two machines' parameter and operation counts agreed to two decimal places.

S8.4. Configuration notes for the after *rows*

The DeepCrack fast-fuse row changes the order of two operations inside the side fusion: the published design concatenates the two side inputs, upsamples, and then convolves, whereas this row convolves at the native side resolution and upsamples the single-channel result. The topology and the parameter count are unchanged—the released weights load into either without a missing or mismatched tensor—but the counted computation is not (Section S7).

The Crack-ShuttleNet row retains repeated multiscale refinement but replaces scalar memory connections with channel-and-spatial gates, adds a boundary auxiliary loss, and uses a lighter body. It is neither the similarly named image-classification ShuffleNet V2 nor the published pavement ShuttleNetV2, whose self-attention and varied sampling scales constitute a different improvement of ShuttleNet (Zhang et al., 2024).

The DTrC-Net row's published Transformer branch, a 256-dimensional eight-head DeiT variant the authors pretrained themselves, was not released. The row uses the public DeiT-tiny-distilled encoder (Touvron et al., 2021) with its ImageNet-1k weights instead, which has 192 dimensions and three heads, so the four fusion inputs narrow accordingly; the ResNet-34 branch keeps its ImageNet-1k weights as released and nothing else in the network changes. Without the substitution the Transformer branch would have trained from random weights, which is not the published model either.

MixerCSeg modified doubles the encoder widths from (16, 32, 64, 128) to (32, 64, 128, 256), widens the decoder from 8 to 64 channels, restricts the mixer's attention to the two deepest stages, drops the dynamic routing and histogram-of-gradients branches, replaces batch normalization with group normalization, fuses the decoder at one quarter of the input resolution and upsamples the logits, simplifies the prediction head, and rebalances the loss toward the foreground class. Under the common protocol the two MixerCSeg configurations differ by −0.60 to +2.33 pixel-ODS points across datasets; at one tenth of the budget the same pair differs by 2.2 to 46.1 points.

Three further rows keep the released network but change its output stage. RINDNet is an edge detector for four kinds of discontinuity; the row uses the single-map edge variant released with it, removes its final sigmoid so that the binary loss of Table 2 can be applied to logits, and replaces its attention loss with that loss. RestorMixer is an image-restoration network whose decoder predicts a residual image; the row keeps its encoder and replaces the decoder with a light segmentation head. MambaVision is a classification backbone; its released segmentation recipe attaches UPerNet heads and loads ImageNet-1k weights, whereas the row attaches a lighter fusion head and trains from random initialization, consistent with the Init. column of Table 4. Each of the three therefore tests its architecture family in a configuration of this study, not the published system. Table S5 collects the code source, license, loaded weights and departures of every row.

S8.5. Reading the model matrix

The accuracy columns of Table 4 are means over seeds 42, 43 and 44 of held-out F1 (Section 2.5). Table S6 gives every seed and, for each cell, the three-seed means of pixel-ODS and of average precision. The Mean column takes the source-disjoint partition of each dataset that provides one, so its CrackMap entry is the rebuild of Section 2.2 rather than the image-level partition distributed with this study's archive copy, and its Crack500 entry is the parent-disjoint column. Accuracy columns are from the common 50-epoch protocol of Section 2.3, and the operation counts are those of Section 2.5. Params and GFLOPs are not bolded because a lower value is preferable there.

RINDNet's two later seeds did not converge on DeepCrack and CrackMap: their validation foreground IoU never exceeded 13.5 in fifty epochs, against 71.5 at the end of the seed-42 run on DeepCrack, with configuration and code unchanged. Its mean is reported as measured, because the final-epoch rule was fixed before any run and excluding runs after seeing them would be a selection on the outcome. Per seed, RINDNet's seed-42 run scores 72.9 to 85.3 on the four Mean columns; seeds 43 and 44 score 16.0 to 23.5 on DeepCrack and CrackMap, 50.8 and 78.4 on CamCrack789, and seed 44 scores 63.3 on parent-disjoint Crack500.

MixerCSeg has a verified *faithful* implementation and a local *modified* variant. Their four-dataset means differ by 0.54 points under the common budget, against 19.8 pixel-ODS points on seed 42 at one tenth of it: the edits chiefly change convergence speed here. Because several settings change together, this pair identifies no individual edit's effect.

Table S5. Where each configuration of Table 4 comes from: the code it was built from, the license of that code, the weights loaded before crack training, and how the trained network departs from the published model.

Model	Code	License	Weights loaded	Departures from the published model
ConvNeXt-XLarge	Framework	Apache-2.0	ImageNet-21k backbone	UPerNet head; 512 by 512 input; common optimizer instead of the layer-decay recipe
U-Net	Framework	Apache-2.0	none	framework variant (batch norm, interpolation upsampling); no auxiliary head
Crack-ShuttleNet[a]	From paper	none released	none	gated instead of scalar memory connections; boundary loss; 9.7 M instead of 58.3 M parameters; RGB instead of 2D+3D input; cross-entropy and Dice instead of focal loss
HRNet-W48	Framework	Apache-2.0	ADE20K segmentor except output layer	two-class output layer
CarNet	Authors' release	MIT	none	released source unchanged; BCE and Dice instead of BCE; input scaled to [-1,1]
DeepLabV3+ (R50)	Framework	Apache-2.0	ADE20K segmentor except output layer	framework variant (deep stem, output stride 8, auxiliary head)
RINDNet[a]	Authors' release	none	ImageNet-1k ResNet-50	single-map edge variant used as a segmentor; final sigmoid removed; BCE and Dice instead of the attention loss
TANGO	This study	this study	none	testbed of this study (Section 2.4)
DeepCrack fast-fuse[a]	Authors' release	none	none	side-fusion order (convolve, then upsample). *Check:* published order trained under the same protocol (Supplementary Section S7)
SegFormer-B5	Framework	Apache-2.0	ADE20K segmentor except output layer	two-class output layer
DTrC-Net[a]	Definition only	none	ImageNet-1k ResNet-34 and DeiT-tiny	unreleased self-pretrained 256-d Transformer branch replaced by public DeiT-tiny (192-d); network run at 256 by 256
RestorMixer[a]	Authors' release	MIT	none	restoration decoder replaced by a segmentation head
MixerCSeg modified[a]	Authors' release	MIT	none	wider encoder and decoder; attention in two stages; no routing or HoG branch; group normalization; low-resolution fusion and simplified prediction head; foreground-weighted loss
MixerCSeg faithful	Authors' release	MIT	none	36 HoG bins, as in the released code (the released weights use 180). *Check:* layer-by-layer agreement with the released code on the released weights (Supplementary Section S8)
MambaVision[a]	Authors' release	NVIDIA Source Code License	none (the released segmentation recipe loads ImageNet-1k)	UPerNet heads replaced by a lighter fusion head
SCSegamba	Authors' release	Apache-2.0	none	BCE and Dice weighted 1:5 (released default 0.83:0.17)

Note: Framework: the implementation in mmsegmentation or mmpretrain. Definition only: the authors released the network definition without training code, weights or a license. From paper: no code was released, so the row was implemented from the paper. [a]Marked *after* in Table 4: a locally derived configuration of the cited design. Settings common to the protocol (loss weights, precision, normalization) are in Table 2; every row was rebuilt from the configuration dumped by its training run, and its parameter count matches Table 4.

Substituting the distributed Crack500 column in the Mean changes every model's mean by at most 0.69 points apart from RINDNet's 0.93. It exchanges two adjacent rows in each of two groups, DeepLabV3+ (R50) with HRNet-W48, whose parent-disjoint means differ by 0.04 points, and MambaVision with SCSegamba; no argument in the main text rests on either order. ConvNeXt-XLarge's margin over Crack-ShuttleNet in the capacity comparison grows from 0.81 to 0.87, so the distributed column would make the capacity payoff look slightly larger than the parent-disjoint one does.

S8.6. Validation trajectories and checkpoint choice

Fig. 1 reports what that budget bought, and it is a property of the runs behind Table 4 rather than a separate experiment: validation scores were logged once per epoch by the runs themselves. No cell was still improving when the budget expired. Over all 80 model-dataset cells the best validation epoch precedes the last, and only ten fall inside the final ten epochs, so the fixed budget starved no configuration. The reported epoch is also a typical member of the plateau it sits on, a median of 0.01 points from the mean of its own last ten epochs, whose spread over those epochs is 0.33 points.

Selecting the best validation epoch raises the validation scores of the cells by a median of 1.44 points, and most of that is selection on the same data: the maximum of fifty draws from a plateau with a 0.33-point spread is expected to exceed the plateau mean by roughly that much. What remains is measured on the test sets, which played no part in the choice. With each checkpoint's threshold also chosen on validation data, the validation-selected checkpoint exceeds the final epoch in test F1 by a median of 0.64 points on distributed Crack500, 0.56 on parent-disjoint Crack500, 0.20 on CamCrack789 and 0.42 on CrackMap, and is higher in 53 of the 64 cells. The largest gains are MambaVision's on the two Crack500 partitions, 2.73 and 4.04 points, and SCSegamba's 11.32 on CrackMap, the overfitting cell described below. The gain is small for the typical model but uneven across models, so it reorders each column: 12, 12, 9 and 13 of the sixteen models change position, by up to 9 places on CrackMap. The final epoch is reported because it is fixed before any result is seen and is one rule for every model and dataset, DeepCrack included; Table 4 therefore understates most configurations by a fraction of a point and a few by more. The threshold is a different matter and is chosen per model, because pixel-ODS is defined as the maximum over a threshold sweep (Section 2.5); no comparable convention exists for checkpoints, and no unbiased one is available.

Three cells are exceptions that none of the above describes, and they are highlighted in Fig. 1 rather than averaged away. MixerCSeg faithful collapses to an all-background output on

Table S6. Three-seed matrix behind main-text Table 4. Upper block: held-out F1 of seeds 42, 43, 44, in that order. Lower block: three-seed means of pixel-ODS, whose threshold is selected on the test set and is therefore an upper bound, and of average precision, which uses no threshold.

Model	DeepCrack	Crack500 dist.	Crack500 p-disj.	CamCrack789	CrackMap
Held-out F1, seed by seed					
CNN-based					
ConvNeXt-XLarge	84.7/84.7/84.2	79.4/79.3/79.4	78.9/78.8/79.1	86.4/86.5/86.5	77.0/76.5/76.3
Crack-ShuttleNet	83.4/83.9/83.6	78.8/78.3/78.9	78.5/78.3/78.6	86.0/86.1/85.7	75.2/74.5/76.1
U-Net	84.3/82.4/84.1	78.2/78.0/77.9	78.1/78.0/77.9	85.5/85.4/85.3	74.5/76.4/75.5
DeepLabV3+ (R50)	83.0/83.0/84.3	77.8/77.3/77.4	78.1/77.8/77.6	85.3/85.4/85.1	73.3/74.5/74.6
HRNet-W48	83.4/82.9/83.7	77.7/77.7/77.8	77.9/77.7/77.9	85.9/85.2/85.4	74.1/74.3/73.3
CarNet	83.2/83.9/83.8	78.1/78.2/78.2	77.7/77.6/77.8	84.7/85.0/84.7	74.6/68.4/74.7
TANGO testbed	84.9/83.7/84.1	76.9/77.0/77.1	76.9/77.1/77.1	84.6/84.6/84.3	71.7/71.8/72.3
DeepCrack fast-fuse	82.5/82.8/82.8	75.2/75.0/75.2	75.1/75.3/74.8	83.4/83.0/83.3	70.8/70.7/70.4
RINDNet	83.5/23.5/22.5	76.9/77.5/75.0	77.6/77.3/63.3	85.3/50.8/78.4	72.9/16.0/19.8
Transformer-based					
SegFormer-B5	84.5/83.1/83.4	78.4/78.5/78.4	78.7/78.4/78.5	86.1/85.9/85.9	74.8/72.6/73.2
DTrC-Net	83.6/82.9/83.5	77.2/77.7/77.6	77.5/77.4/77.7	84.9/84.6/85.0	72.5/74.8/73.0
Mamba-based					
RestorMixer	83.7/83.6/82.9	74.8/77.6/78.3	75.0/74.8/72.6	85.5/85.9/85.7	74.6/73.9/73.8
MixerCSeg modified	82.3/81.1/82.1	77.3/77.0/77.3	76.9/77.4/77.0	83.5/83.8/83.9	72.3/71.7/71.4
MixerCSeg faithful	82.9/82.3/82.5	77.0/76.9/77.0	76.6/77.1/77.1	81.0/82.3/81.6	70.4/72.6/70.4
MambaVision	80.5/81.7/75.5	69.3/72.7/74.4	71.3/76.2/76.2	84.4/84.5/82.9	73.1/70.7/72.6
SCSegamba	79.7/80.9/80.7	78.2/78.0/78.4	77.0/78.0/78.1	84.5/84.8/85.0	58.6/65.0/69.8
Pixel-ODS / average precision, three-seed means					
CNN-based					
ConvNeXt-XLarge	84.6 / 89.2	79.4 / 81.6	78.9 / 80.0	86.5 / 92.3	77.0 / 82.9
Crack-ShuttleNet	83.7 / 87.9	78.8 / 83.9	78.5 / 84.0	86.0 / 91.0	75.4 / 76.9
U-Net	83.7 / 90.3	78.1 / 85.5	78.0 / 84.9	85.4 / 93.1	76.1 / 82.9
DeepLabV3+ (R50)	83.5 / 89.6	77.5 / 82.4	77.8 / 81.7	85.3 / 91.4	74.4 / 78.3
HRNet-W48	83.4 / 89.8	77.8 / 82.7	77.8 / 81.8	85.5 / 92.3	74.1 / 77.8
CarNet	83.7 / 85.7	78.2 / 80.1	77.7 / 79.2	84.9 / 87.2	72.6 / 71.9
TANGO testbed	84.3 / 87.0	77.0 / 79.6	77.1 / 79.1	84.5 / 88.4	72.2 / 72.5
DeepCrack fast-fuse	82.7 / 87.5	75.1 / 77.8	75.1 / 76.5	83.2 / 89.7	72.0 / 77.4
RINDNet	43.2 / 35.6	76.6 / 71.1	72.8 / 65.4	71.5 / 61.2	36.4 / 27.4
Transformer-based					
SegFormer-B5	83.7 / 89.3	78.5 / 81.6	78.5 / 80.3	86.0 / 92.7	74.4 / 77.7
DTrC-Net	83.4 / 81.2	77.6 / 79.9	77.5 / 79.7	84.9 / 84.0	73.5 / 67.4
Mamba-based					
RestorMixer	83.5 / 84.2	76.9 / 80.2	74.3 / 77.3	85.7 / 88.2	74.6 / 76.4
MixerCSeg modified	81.8 / 85.7	77.2 / 82.3	77.1 / 81.6	83.8 / 89.1	72.1 / 74.4
MixerCSeg faithful	82.6 / 85.6	77.0 / 83.3	76.9 / 83.1	81.7 / 88.1	72.2 / 70.3
MambaVision	79.5 / 82.9	72.3 / 74.9	74.6 / 78.7	84.0 / 87.2	73.0 / 76.1
SCSegamba	80.5 / 78.4	78.2 / 81.9	77.7 / 80.9	84.8 / 85.0	64.9 / 58.2

DeepCrack and on CrackMap, holding a validation score of exactly zero for the last ten epochs; Section 3.2 reports what its score becomes once the threshold is swept. SCSegamba peaks at epoch 17 on CrackMap and loses 18.36 points of validation foreground IoU by the last one, which is overfitting to 84 training images and the only cell in the matrix where the longer budget costs more than a point.

S8.7. Test-set sampling uncertainty

How much of this ordering the test sets themselves resolve is a separate question. Resampling each test set of seed 42 with replacement 10,000 times, by source photograph where the dataset is tiled, and selecting each model's threshold again inside every resample gives a sampling interval per cell whose median half-width is 1.84 points on DeepCrack, 2.28 on parent-disjoint Crack500, 1.34 on CamCrack789 and 3.44 on CrackMap. Holding each model's threshold fixed at its full-set value instead changes these medians by at most 0.11 points, so a threshold fitted to the test set does not make the intervals look narrower than they are. Differencing two models on each resample, no adjacent pair in the sixteen-model ranking separates from zero on DeepCrack or parent-disjoint Crack500, one of fifteen does on CrackMap and two of fifteen do on CamCrack789; wider gaps do separate, 59 of the 120 model pairs on DeepCrack, 77 on parent-disjoint Crack500, 90 on CamCrack789 and 43 on CrackMap. These intervals condition on one trained run per model and so omit training variation, which Table S6 shows separately. The matrix therefore supports statements about groups of configurations rather than neighbouring positions, and the attributions in this study rest on the matched contrasts of Sections 4.2–4.4.

S9. Reporting practice in the cited crack-segmentation literature

Section 2.1 of the main text justifies the six choices by how often published comparisons leave them unstated or unmatched. The count rests on a coding of every crack-segmentation paper cited in the manuscript that proposes a deep pixel-level crack segmentation model and reports quantitative test results, 29 papers once one is excluded because its pixel masks come from a non-learned threshold rather than a trained model. All 29 were read in full, from the published version, an open preprint, or the authors' thesis, so every count below covers the whole sample rather than the papers that happened to be accessible.

Each paper was coded on ten fields with a fixed set of values, every value backed by a short quotation and its location in the text; anything the paper does not state was coded as not reported, without inference from released code or from other papers. Rulings for ambiguous cases were fixed before the counts were taken: a split counts as source-disjoint only when the paper says so, when training and test images come from different datasets or acquisitions, or when it splits whole photographs before tiling; a binarization rule counts as stated only for the metric it is written for, so a paper that reports a swept ODS alongside an F1 or IoU of unstated origin counts as not stating the latter. The protocol, the per-paper coding with its quotations, and every change made when the coding was merged are released with the data of this study.

Table S7 gives the counts. They describe how much of a comparison a reader must reconstruct before reading it, not the quality of any individual study, and no paper is named.

Table S7. Reporting practice in the crack-segmentation papers read in full (all 29 papers of a pre-specified sample; Section S9).

Factor	Not reported or not matched	Papers	Breakdown
Partition	Source-photograph assignment not established	11 of 14	Denominator excludes datasets of whole photographs, where the unit is moot
Threshold	Binarization of the reported masks not stated	16 of 29	3 of these also report ODS; 4 fixed, 8 swept, 1 other
Checkpoint	Rule not stated, or selected on test data	19 of 29	15 not stated, 4 on test data
Budget	Epochs or iterations not stated	4 of 29	Batch size not stated in 7
Initialization	Starting weights not stated	12 of 29	10 pretrained, 7 from scratch
Seed variation	Single run, no dispersion across runs	29 of 29	
Capacity	No parameter or operation count	13 of 29	
Comparison	Baselines not stated as retrained under the same protocol	7 of 29	Copied, mixed, or not stated

S10. Native-resolution processing

The common protocol resizes every image to 512 by 512 pixels (Section 2.3), which stretches CamCrack789's 640 by 480 images and upsamples CrackMap's 256 by 256 ones. To test whether this choice changes accuracy, geometry or rankings, six representative models were retrained at native resolution with seeds 42, 43 and 44 under an otherwise unchanged protocol: CamCrack789 is padded to 640 by 640 with its aspect ratio kept and the padding ignored in the loss and the metrics, and CrackMap is used at 256 by 256. Both arms are scored against native-resolution ground truth at a threshold selected on native-resolution validation images; the resized arm uses the matrix checkpoints of the same seed, with its probability maps resized back before scoring. Unlike the other contrasts, whose two arms share one machine, the native runs were trained on a second machine with a different software stack; all scoring was done on one machine.

Table S8 gives the result. On CamCrack789 native processing scores higher in all eighteen model-and-seed cells, by 0.47 to 1.46 points, narrows the over-estimate of crack width in all eighteen, and leaves the ranking almost unchanged, with a rank correlation of 0.94 to 1.00 between the arms on each seed. With three seeds and twelve contrasts only ConvNeXt-XLarge's gain survives Holm correction, so the evidence for a general gain is its consistency rather than any single interval. On CrackMap the direction depends on the model: U-Net gains on every seed, while ConvNeXt-XLarge and MixerCSeg faithful lose on every seed. SCSegamba gains 8.40 points on average, but by 13.89, 8.71 and 2.62 on the three seeds, because its resized runs vary widely; its low CrackMap entry in Table 4 is therefore partly a product of upsampling. Rankings change for two to six of the six models (rank correlation 0.66 to 0.77), and predicted widths are larger at native resolution in all eighteen cells. The fraction of long cracks broken into pieces changed in no consistent direction on either dataset, and no contrast on it survived correction; CrackMap's 24 test images contain only 32 long cracks, so one crack moves that fraction by 3.1 points. The resizing choice of the common protocol thus slightly lowers CamCrack789 scores without reordering the models, and on CrackMap it is a source of ranking variation in its own right, which the reporting checklist's resizing item asks authors to state.

Table S8. Native-resolution training against the 512 by 512 protocol, six models, seeds 42, 43, 44. Held-out F1 and width bias are three-seed means on native-resolution ground truth; the change is native minus resized, paired by seed, with its 95% paired *t* interval. Bold marks a change that survives Holm correction across the twelve contrasts. Width bias is predicted minus true mean crack width in pixels, the median over images.

Model	F1 resized	F1 native	Change [95% CI]	Width bias resized	Width bias native
CamCrack789 (640 by 480)					
ConvNeXt-XLarge	86.3	87.4	**+1.11 [+1.03, +1.19]**	0.65	0.30
SegFormer-B5	85.8	86.7	+0.86 [+0.40, +1.32]	1.07	0.49
U-Net	85.5	86.2	+0.73 [−0.06, +1.52]	0.53	0.31
SCSegamba	84.5	85.5	+1.04 [+0.36, +1.72]	0.71	0.28
TANGO testbed	84.3	85.4	+1.15 [+0.49, +1.82]	0.62	0.46
MixerCSeg faithful	81.5	82.2	+0.71 [+0.07, +1.34]	1.24	0.94
CrackMap (256 by 256)					
U-Net	75.7	77.0	+1.32 [+0.48, +2.17]	1.52	1.89
ConvNeXt-XLarge	76.6	75.3	−1.28 [−2.51, −0.04]	1.72	2.26
SegFormer-B5	73.7	73.7	+0.03 [−3.53, +3.59]	1.27	2.15
SCSegamba	64.7	73.1	+8.40 [−5.61, +22.42]	2.05	3.44
TANGO testbed	71.9	71.4	−0.49 [−1.37, +0.40]	2.02	2.54
MixerCSeg faithful	71.5	69.4	−2.11 [−2.74, −1.47]	2.74	3.60

S11. Crack geometry metrics

Pixel F1 rewards overlap and is indifferent to whether a crack comes out in one piece. The geometry metrics of Table 6 and Section 4.4 are computed on binary masks at the same held-out threshold as the F1, image by image at the evaluation resolution, and pooled over the test set. *Centerline Dice* is the clDice of Shit et al. (2021). A ground-truth crack is an eight-connected component whose skeleton has at least 20 pixels; it is *broken* when predicted components within a three-pixel corridor around it number two or more, and *missed* when none falls in the corridor. *Thin-crack recall* is the share of ground-truth skeleton pixels whose local width, twice the distance-transform value, is below 3 pixels and which lie within 2 pixels of the prediction; widths of 3 to 7 and above 7 pixels give the medium and wide strata. *Width bias* is the predicted mean crack width, area over skeleton length, minus the same quantity for the ground truth, the median over images. *Spurious components* are predicted components farther than five pixels from any ground-truth crack, counted per image. Endpoint and junction counts after pruning skeleton spurs of at most ten pixels, length error, and stratification by local contrast are stored with the release.

Across the sixteen-model matrix at seed 42, the geometry metrics order models differently from F1: the rank correlation between F1 and centerline Dice, negative broken fraction or thin-crack recall ranges from 0.35 to 0.82 across the four datasets. Thin cracks are recalled less often than wide ones by every median: 0.70 against 0.93 on DeepCrack, 0.62 against 0.88 on parent-disjoint Crack500, 0.87 against 0.97 on CamCrack789 and 0.53 against 0.85 on CrackMap. Every model overestimates crack width on DeepCrack, Crack500 and CamCrack789, by 0.35 to 5.75 pixels; on CrackMap, whose 256-pixel images are upsampled for training, the bias runs from −4.22 to +8.98 pixels and three models underestimate. Tables S9 and S10 give every run behind Table 6.

Table S9. Every run behind Table 6: per-seed scores of the from-scratch reference and of each single-change arm, the paired mean change with its 95% t interval, and the unadjusted and Holm-adjusted p values (ten contrasts per metric).

Run	**Per-seed score (5 seeds)**	**Change [95% CI]**	p	**Holm** p
Held-out F1				
DeepCrack, reference	84.84 / 84.39 / 84.15 / 84.39 / 84.78			
Remove propagation	84.57 / 84.82 / 85.27 / 84.19 / 84.08	+0.08 [−0.80, +0.96]	0.823	1.000
Remove orientation sup.	84.29 / 84.35 / 84.90 / 84.65 / 84.27	−0.02 [−0.69, +0.66]	0.946	1.000
Remove clDice loss	85.17 / 84.52 / 84.69 / 84.83 / 85.00	+0.33 [+0.13, +0.53]	0.010	0.087
Uniform steering	84.87 / 83.79 / 84.50 / 84.83 / 84.81	+0.05 [−0.45, +0.55]	0.788	1.000
Add IN-1k init.	84.81 / 84.44 / 84.46 / 84.51 / 84.83	+0.10 [−0.06, +0.26]	0.149	1.000
Crack500 (p-disj.), reference	77.28 / 77.10 / 77.11 / 77.11 / 77.19			
Remove propagation	77.25 / 77.19 / 77.08 / 77.23 / 77.24	+0.04 [−0.05, +0.12]	0.319	1.000
Remove orientation sup.	76.91 / 76.83 / 76.92 / 77.13 / 77.36	−0.13 [−0.40, +0.14]	0.252	1.000
Remove clDice loss	77.39 / 77.37 / 77.25 / 77.39 / 77.26	+0.17 [+0.05, +0.29]	0.016	0.127
Uniform steering	77.20 / 77.17 / 77.06 / 77.06 / 77.52	+0.04 [−0.16, +0.25]	0.580	1.000
Add IN-1k init.	78.44 / 78.38 / 78.50 / 78.65 / 78.61	+1.36 [+1.18, +1.53]	< 0.001	< 0.001
Centerline Dice				
DeepCrack, reference	88.13 / 88.45 / 87.97 / 88.46 / 88.34			
Remove propagation	88.81 / 88.56 / 88.81 / 88.14 / 87.90	+0.18 [−0.54, +0.89]	0.534	1.000
Remove orientation sup.	87.61 / 87.93 / 87.87 / 88.23 / 88.26	−0.29 [−0.56, −0.02]	0.041	0.204
Remove clDice loss	87.95 / 88.00 / 87.42 / 87.55 / 88.15	−0.46 [−0.83, −0.09]	0.027	0.161
Uniform steering	88.62 / 88.05 / 87.96 / 88.53 / 88.41	+0.04 [−0.34, +0.43]	0.764	1.000
Add IN-1k init.	89.09 / 88.81 / 89.00 / 88.67 / 89.35	+0.71 [+0.22, +1.21]	0.016	0.112
Crack500 (p-disj.), reference	80.63 / 80.21 / 80.23 / 80.20 / 80.35			
Remove propagation	81.31 / 80.73 / 80.57 / 80.62 / 80.94	+0.51 [+0.34, +0.68]	0.001	0.010
Remove orientation sup.	80.26 / 80.24 / 80.06 / 80.65 / 80.62	+0.04 [−0.37, +0.46]	0.781	1.000
Remove clDice loss	78.71 / 79.39 / 78.99 / 79.28 / 78.95	−1.26 [−1.80, −0.71]	0.003	0.024
Uniform steering	80.42 / 80.36 / 80.30 / 80.27 / 80.74	+0.09 [−0.18, +0.36]	0.395	1.000
Add IN-1k init.	82.22 / 81.77 / 81.63 / 82.03 / 82.30	+1.67 [+1.39, +1.94]	< 0.001	< 0.001

Table S10. Every run behind Table 6, continued, with the spurious-component count: per-seed scores of the from-scratch reference and of each single-change arm, the paired mean change with its 95% t interval, and the unadjusted and Holm-adjusted p values (ten contrasts per metric).

Run	**Per-seed score (5 seeds)**	**Change [95% CI]**	p	**Holm** p
Broken long cracks (%)				
DeepCrack, reference	19.22 / 18.16 / 18.16 / 15.68 / 17.81			
Remove propagation	21.70 / 21.11 / 19.22 / 17.45 / 19.46	+1.98 [+1.06, +2.90]	0.004	0.031
Remove orientation sup.	22.76 / 19.46 / 18.99 / 20.40 / 20.52	+2.62 [+0.63, +4.60]	0.022	0.126
Remove clDice loss	24.06 / 24.29 / 22.52 / 22.05 / 24.29	+5.64 [+4.43, +6.84]	< 0.001	0.002
Uniform steering	20.17 / 18.04 / 18.40 / 19.58 / 20.75	+1.58 [−0.60, +3.76]	0.114	0.343
Add IN-1k init.	17.22 / 18.04 / 17.45 / 16.75 / 17.57	−0.40 [−1.78, +0.97]	0.464	0.928
Crack500 (p-disj.), reference	7.54 / 7.81 / 8.77 / 7.46 / 8.25			
Remove propagation	9.12 / 9.82 / 9.30 / 8.77 / 9.39	+1.32 [+0.63, +2.00]	0.006	0.042
Remove orientation sup.	9.47 / 9.21 / 9.56 / 9.65 / 8.51	+1.32 [+0.33, +2.31]	0.021	0.126
Remove clDice loss	11.67 / 11.23 / 12.98 / 11.23 / 10.18	+3.49 [+2.34, +4.64]	0.001	0.010
Uniform steering	9.39 / 7.98 / 8.42 / 7.63 / 7.46	+0.21 [−1.03, +1.45]	0.662	0.928
Add IN-1k init.	6.75 / 7.89 / 7.37 / 6.14 / 6.67	−1.00 [−1.84, −0.16]	0.030	0.126
Thin-crack recall (%)				
DeepCrack, reference	66.40 / 67.60 / 65.98 / 68.51 / 68.22			
Remove propagation	66.83 / 67.69 / 66.50 / 67.76 / 66.77	−0.24 [−1.28, +0.81]	0.564	1.000
Remove orientation sup.	65.41 / 66.69 / 65.48 / 67.01 / 67.33	−0.96 [−1.41, −0.52]	0.004	0.027
Remove clDice loss	68.34 / 70.42 / 68.49 / 70.41 / 70.49	+2.29 [+1.80, +2.77]	< 0.001	0.002
Uniform steering	67.10 / 67.25 / 66.76 / 67.20 / 67.28	−0.23 [−1.40, +0.94]	0.616	1.000
Add IN-1k init.	74.55 / 73.91 / 76.56 / 74.25 / 74.58	+7.42 [+4.97, +9.88]	0.001	0.009
Crack500 (p-disj.), reference	65.35 / 63.68 / 64.15 / 64.05 / 63.79			
Remove propagation	63.33 / 62.65 / 62.92 / 62.44 / 63.75	−1.19 [−2.11, −0.27]	0.023	0.116
Remove orientation sup.	64.60 / 64.12 / 63.15 / 63.30 / 63.59	−0.45 [−1.17, +0.27]	0.157	0.626
Remove clDice loss	63.95 / 61.78 / 62.69 / 62.09 / 62.72	−1.56 [−2.02, −1.10]	< 0.001	0.006
Uniform steering	63.47 / 62.65 / 63.46 / 64.39 / 63.72	−0.67 [−1.74, +0.40]	0.159	0.626
Add IN-1k init.	65.95 / 64.33 / 65.24 / 66.01 / 64.49	+1.00 [+0.29, +1.71]	0.017	0.103
Spurious components per image				
DeepCrack, reference	1.20 / 1.14 / 0.97 / 1.24 / 1.24			
Remove propagation	0.92 / 1.09 / 1.00 / 0.89 / 1.03	−0.17 [−0.37, +0.02]	0.069	0.277
Remove orientation sup.	1.34 / 1.35 / 1.11 / 1.32 / 1.55	+0.18 [+0.06, +0.29]	0.012	0.069
Remove clDice loss	1.72 / 2.04 / 1.75 / 1.88 / 2.12	+0.75 [+0.54, +0.95]	< 0.001	0.005
Uniform steering	1.15 / 1.06 / 1.21 / 1.35 / 1.23	+0.04 [−0.12, +0.20]	0.481	0.764
Add IN-1k init.	1.00 / 0.74 / 0.87 / 0.77 / 0.97	−0.29 [−0.46, −0.11]	0.011	0.069
Crack500 (p-disj.), reference	0.28 / 0.28 / 0.28 / 0.31 / 0.29			
Remove propagation	0.26 / 0.26 / 0.24 / 0.26 / 0.24	−0.04 [−0.06, −0.01]	0.009	0.066
Remove orientation sup.	0.34 / 0.30 / 0.32 / 0.35 / 0.32	+0.04 [+0.02, +0.06]	0.006	0.045
Remove clDice loss	0.50 / 0.43 / 0.42 / 0.49 / 0.45	+0.17 [+0.13, +0.21]	< 0.001	0.003
Uniform steering	0.30 / 0.32 / 0.31 / 0.28 / 0.28	+0.01 [−0.02, +0.05]	0.382	0.764
Add IN-1k init.	0.29 / 0.20 / 0.27 / 0.25 / 0.27	−0.03 [−0.08, +0.01]	0.126	0.379

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